GRADS April 2026

Oraly presented in the *Computing, Data, Modeling, and Robotics; Technology, Policy, Health and Human Systems* section.

Help with slides from Justin Kilb.

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Modeling Logistics in a Contested Environment

Partnered with Columbia and funded by the Office of Naval Research

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April 03, 2026

OPERATIONS RESEARCH WITH ENGINEERING

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Our Task

Purpose of this work

- The Office of Naval Research (ONR) has requested that we predict Red behavior and effects within a Marine logistics network.
- A logistics network coordinates the movement of supplies to meet demand. These operations are crucial because:
  - Conventional military units at all echelons and of all types have a relatively short life expectancy if they are not constantly resupplied.

This then begs the question:

Where is Blue most vulnerable to Red actions, and how can we redesign the logistics network to protect Blue against difficult-to-predict outcomes?

We postulate the following:

- A representation of Red interference as adversarial, logistics-driven disruption that is **poorly understood**.
- An assumption that small Red actions create **disproportionate effects to Blue's logistics.**
- An iterative loop between Blue and Red, modeled after network interdiction frameworks and conceptually similar to generative adversarial networks, in which each side adapts to the other's actions.

Building a Robust Blue/Red Model

Problem Statement

- Current planning does not account for **enemy** disruption
- Existing tools cannot show vulnerabilities under attack
- Need: a way to stress-test supply routes against enemy actions

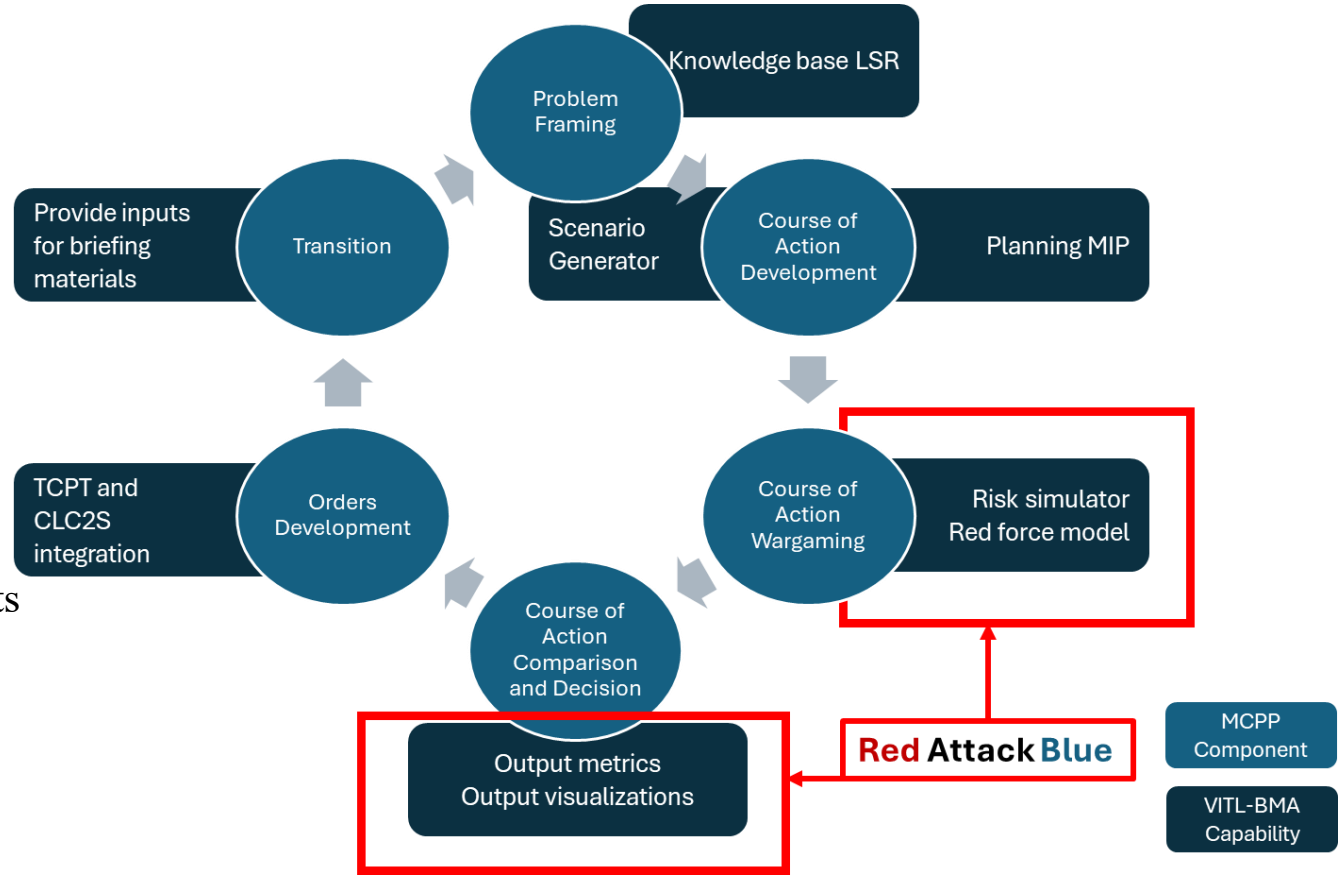
Objective: Red Attack Blue

- Add a **Red Force** module into current planning tools
- See logistics through the enemy's POV to reveal weak points
- Give planners options to strengthen supply routes

How It Fits in the Marine Corps Planning Process (MCPP)

- Mirrors wargaming: **Blue** moves → **Red** interferes → **Blue** adapts
- Iterative cycle updates plans as new info arrives
- Results show likely disruptions and how **Blue** can adjust

VITL-BMA in MCPP



Red Formulation

Red seeks to maximally disrupt Blue's plan such that:

- The first objective term incentivizes Red disruption on heavily used Blue links.
- The log barrier terms prevent Red from reaching their attack budget.
- Incorporates Red *reach* to model given red intel

Assumptions We Address

- **Red** is adaptive, not bound by doctrine
- **Red** actions uncertain; capabilities and locations unknown
- Model integrates intel without fixed behavior assumptions

Red Model (Stress Test)

- Starts from same plan, then simulates **attacks**
- Maximizes **disruption** while conserving **Red** capability
- Identifies vulnerabilities → triggers re-optimization

Objective function

Red variable: Red's exponential disruption on Blue link

Red variable: Red's effort to disrupt Blue's flow on link

$$(\mathcal{R}) \quad \max \sum_{\substack{(\ell,\ell')\in\bar{\mathcal{R}}:\\ N_{v\ell\ell' rt}>0}} n_{v\ell\ell' rt} Z_{\ell\ell'} + \epsilon \cdot \underbrace{\sum_{\substack{(\ell,\ell')\in\bar{\mathcal{R}}:\\ N_{v\ell\ell' rt}>0}} \log\left(b_{\ell\ell'} - W_{\ell\ell'}\right)}_{\text{Per-link attack budget}} + \hat{\epsilon} \cdot \underbrace{\log\left(\beta - \sum_{\substack{(\ell,\ell')\in\bar{\mathcal{R}}:\\ N_{v\ell\ell' rt}>0}} W_{\ell\ell'}\right)}_{\text{Total attack budget}}$$

Blue parameter: Amount of commodity flow on the link

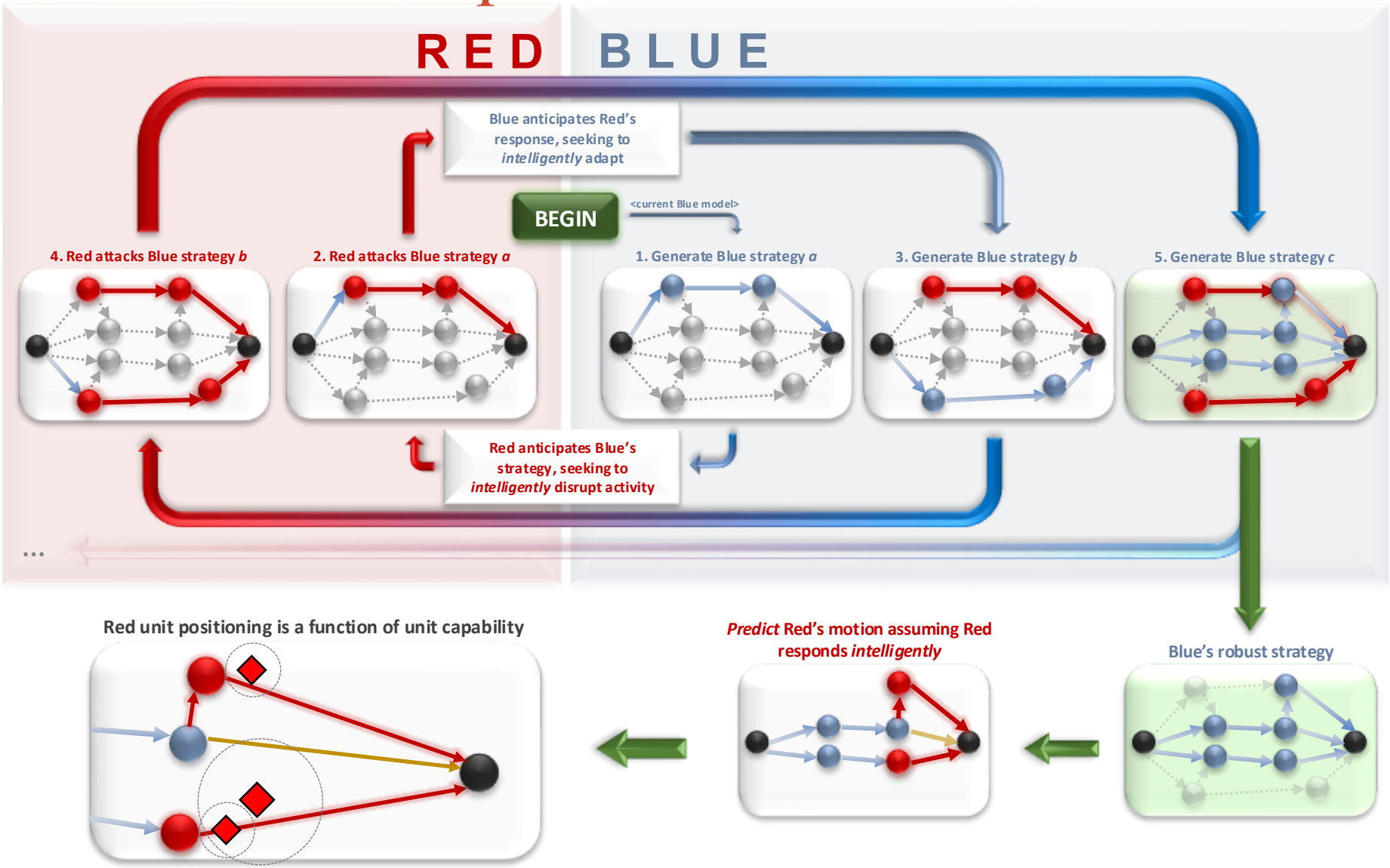
Subject to

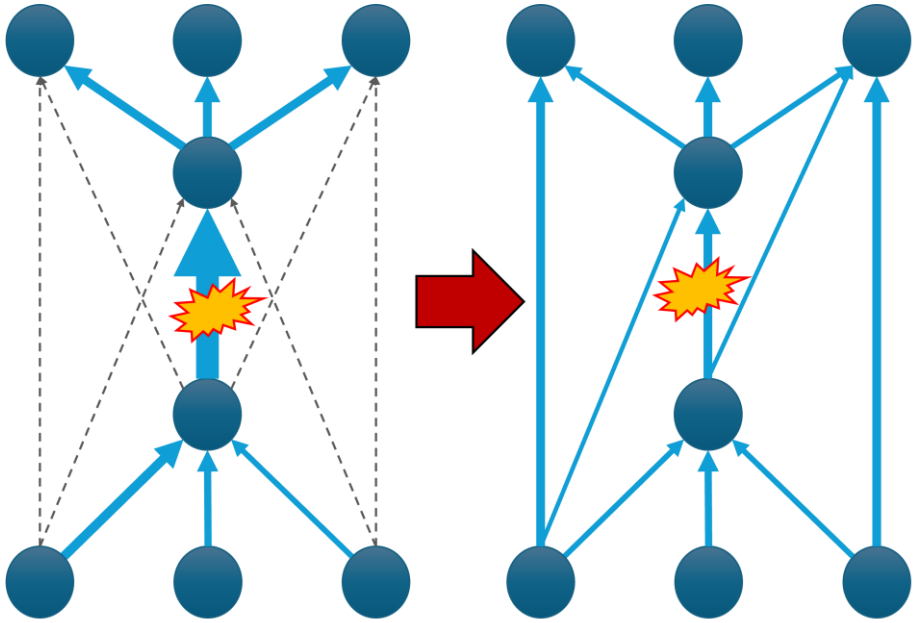
Red's ability to create a disproportionate outcome relative to its budget

$$Z_{\ell\ell'} = e^{W_{\ell\ell'}} - 1 \quad \forall (\ell,\ell') \in \bar{\mathcal{R}} \quad \text{(Effort-to-impact mapping)} \quad (2a)$$

$$0 \le W_{\ell\ell'} \le 1 \quad \forall (\ell,\ell') \in \bar{\mathcal{R}} \quad \text{(Effort domain)} \quad (2b)$$

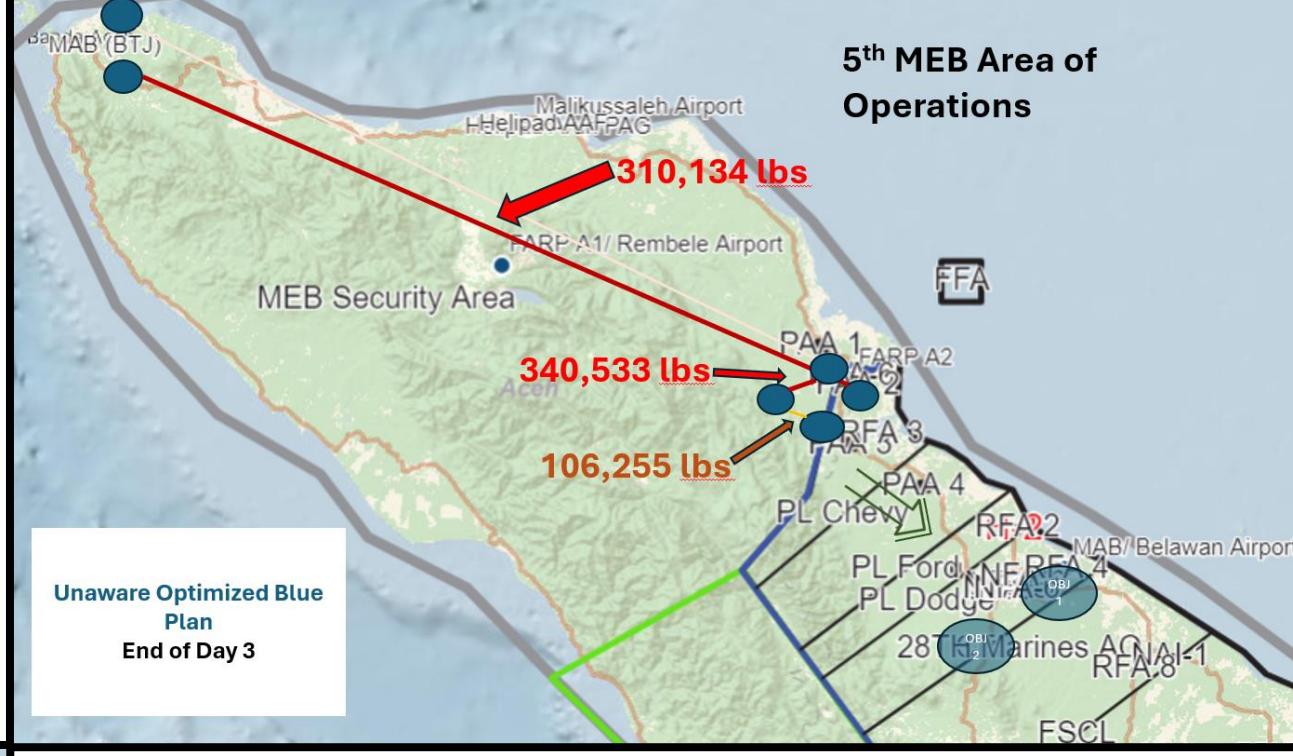
Network interdiction loop

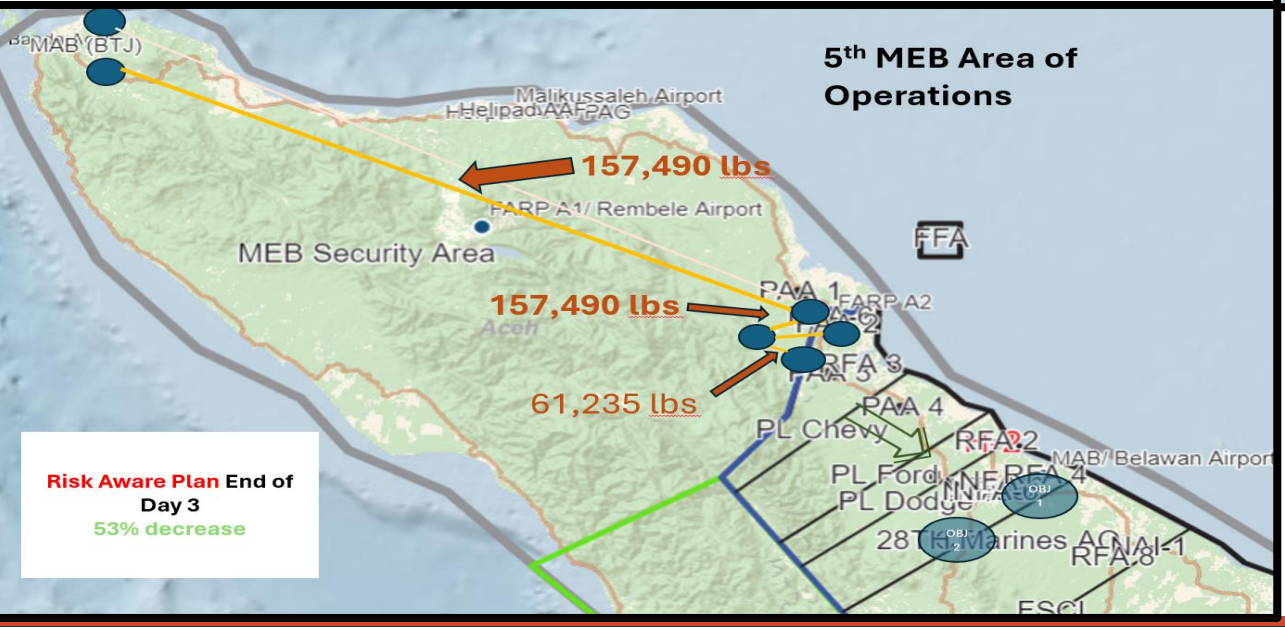




Supply chain is optimized but vulnerable.

Supply chain is robust while still meeting the mission.





Summary

Non-Risk Aware Plan

Demand Met: 86.53%
Max Weight: 340,553lbs

Top 5 Risky Legs

1. 3/28 Staging -FARP 2A
2. 3/13 Staging - PAA2
3. 3/28 Staging -1/28 Area
4. Sultan Airport –Pekanbaru AB
5. 3/28 Staging- PAA2

Risk Aware Plan

Demand Met: 87.56%
Max Weight: 157,490lbs

Top 5 Risky Legs

1. 3/13 Staging- PAA2
2. 3/28 Staging- FARP 2A
3. 3/28 Staging Location
4. Banda Port- Pekanbaru AB
5. 3/28 Staging- 1/28 Support

Red Models

- The team has been tasked with exploring a variety of Red models.
- The plan is to use a variety of simple and some more complex models to "stress test" logistics plans.
- Will explore a wide range of ideas: drones (reusable/disposable), hypersonic missiles, different attack strategies (infiltration vs frontal), uncertainty, etc.
- Idea is that by testing logistics plans across a variety of attacks we can identify weak points in the plan and inform decision makers.

General Challenges

- Need to tailor model inputs to be like what Marine Corps logistics elements would have available to them.
- Be able to a wide range of scenarios.
- Making reasonable assumptions about Red is difficult.
- Giving models more specificity requires more assumptions about Red.

Table of Models

	Robust	Imperfect Robust	Attrition	Position	Hazard
Type	Deterministic	Mix	Deterministic	Deterministic	Deterministic
Modality	Abstract	Abstract	Abstract	Mix	Real
Attacks	Arcs	Arcs	Either	Nodes	Arcs
Implemented	Yes	No	WIP	WIP	No

Summary

- Robust Blue/Red model that focuses on generating resilient logistics plans.
- Developing Red models that seek to attack logistics plans to allow planners more insight in to where they are most vulnerable.

Next Steps

- Focusing on building Red models for the conceivable future.

Thank You

Acknowledgements:

Alexandra Newman: School of Mines
Daniel Bienstock: Columbia University

Caleb Fluker: School of Mines
Justin Kilb: School of Mines
Gabe Hake: School of Mines
Bobby Provine: School of Mines
Brandon Werling: School of Mines
Blake Sisson: Columbia University

Dr. William Krebs: Office of Naval Research: Mathematics, Computer and Information Sciences
Dr. David Phillips III: Program Officer, Mathematical and Resource Optimization, Computational Methods for Decision Making, Applied and Computational Analysis
LTC Reyna: 3rd MARDIV logistician
Vice Admiral Scott Pappano: Principal Military Deputy Assistant Secretary of the Navy
Colonel Jose Gonzalez: Deputy Branch Head of the USMC Logistics Information Technology Branch
Colonel Julian Tsukano: 3rd Marine Division Senior Logistics Officer